

SCOPE OF WORK

Project: Transportation Systems

Client: Federal Highway Administration (FHWA)

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Project Overview

The Federal Highway Administration seeks to better understand real-world travel behavior and microscopic traffic dynamics to support improved traffic modeling, safety initiatives, and intelligent transportation systems. This project combines analysis of the **National Household Travel Survey (NHTS)** and **Next Generation Simulation (NGSIM)** datasets to examine both broad household travel patterns and detailed vehicle-level car-following behavior.

Scope of Work – Tasks Undertaken

1. Identification of Variables Examined in Graphs and Analysis

The team examined variables from both datasets as follows:

From the NHTS Dataset (Household, Person, Vehicle, and Trip files):

- **household_income** (HHFAMINC) – Categorical household income brackets. This variable was selected to understand how income levels influence travel behavior and vehicle ownership.
- **household_size** (HHSIZE) – Total number of people in the household. Selected to analyze the relationship between household composition and trip generation.
- **household_workers** (WRKCOUNT) – Number of workers in the household. Included to examine the impact of employment on daily travel demand.
- **urban / household_location** (URBRUR) – Urban vs. rural classification of the household. Chosen to compare travel patterns between urban and rural areas.
- **vehicle_type** – Type of vehicle owned (car, SUV, pickup truck, etc.). Selected to understand fleet composition and its relation to travel modes.

- **TRIPPURP / WHYTRP1S** – Trip purpose (e.g., home-based work, shopping, social/recreational). Included to identify the primary reasons for travel and modal choices.

From the NGSIM Dataset:

- **trajectory_number** – Unique identifier for each vehicle trajectory. Used to isolate individual vehicle paths for detailed microscopic analysis.
- **Time** – Timestamp in seconds. Selected as the temporal reference for all time-series visualizations.
- **leader_speed(m/s)** – Speed of the preceding (lead) vehicle in meters per second.
- **follower_speed(m/s)** – Speed of the subject (following) vehicle in meters per second.

These NGSIM variables were chosen because they directly capture **car-following dynamics**, a core component of microscopic traffic simulation models used by FHWA.

2. Accurate Explanation of NHTS and NGSIM Variables Included in the Dataset

NHTS Variables:

- **household_income (HHFAMINC)**: Represents the annual income category of the household. It is a categorical variable with coded brackets (e.g., 01 = Less than \$10,000, up to 11 = \$200,000 or more). This variable helps reveal socioeconomic influences on travel demand, vehicle ownership, and mode choice.
- **household_size (HHSIZE)** and **num_adults_hh (NUMADLT)**: Numeric counts of total household members and adults (age 18+). These are critical for normalizing trip rates and understanding how household composition affects daily travel.
- **urban / household_location (URBRUR)**: Binary classification (01=Urban, 02=Rural) based on 2020 TIGER/Line shapefiles. This variable allows differentiation of travel behavior and infrastructure needs between densely populated and rural settings.
- **TRIPPURP / WHYTRP1S**: Summarized trip purpose categories. These derived variables classify each trip (e.g., work, shopping, social) and are essential for understanding trip generation and distribution patterns.

NGSIM Variables:

- **trajectory_number:** A unique integer identifier assigned to each vehicle's complete path through the study corridor. It enables the extraction of individual vehicle trajectories for focused microscopic analysis.
- **Time:** Continuous time stamp recorded in seconds. This serves as the independent variable in all time-series plots, allowing precise observation of speed changes over the duration of the trajectory.
- **leader_speed(m/s)** and **follower_speed(m/s):** Instantaneous speeds of the lead and following vehicles, measured in meters per second. These variables were specifically selected to visualize and analyze real-world car-following behavior, including reaction lags and speed synchronization between consecutive vehicles.

3. Tasks Completed by the Team

- Loaded and cleaned both NHTS and NGSIM datasets using Python (pandas).
- Filtered NGSIM data to a single representative trajectory (Trajectory #1) for focused analysis.
- Created time-series visualizations comparing leader and follower vehicle speeds over time.
- Produced an annotated code document explaining all steps and variable selections.
- Developed a formal summary report linking findings to FHWA's transportation systems goals.